

AWS Solutions Architect Associate (SAA-C03) Syllabus

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Course Overview

This syllabus lists every module of the course in teaching order, with the topics covered in each. It is generated from the course page, so the two cannot disagree.

Module 1: Exam Strategy, Account Setup & Foundations Review

Schedule: Week 1

How the SAA-C03 exam actually scores you (the four domains and their weightings), the question-writing patterns AWS uses (best vs least-bad answers, scenario keywords that signal which service is expected), test-day logistics. Then a foundations review — IAM principals / policies / roles, AWS Organizations + SCPs, Control Tower for multi-account, plus the shared responsibility model. By the end of week 1 every student has done their first 50-question diagnostic exam to establish baseline.

Topics

- SAA-C03 exam structure and domain weighting
- Question patterns — best, least-bad, scenario keywords
- Test-day logistics — Pearson VUE
- IAM principals, policies, roles in depth
- AWS Organizations and SCPs
- AWS Control Tower
- First diagnostic exam

Module 2: Core Services Refresher — EC2, S3, VPC, IAM

Schedule: Week 1

SAA-C03 is a design exam, not a services exam — but it assumes you already know what each service does, and a candidate who is still learning what an internet gateway is cannot evaluate an architecture that contains one. This week levels that floor in the console and the CLI rather than in slides.

You launch instances and understand the families and pricing models, work through S3 buckets, versioning and lifecycle, build a small VPC by hand, and set up IAM users, groups and roles. Anyone arriving from the AWS Cloud Computing course can treat this as revision; anyone arriving from a Cloud Practitioner background will need it.

Topics

- AWS global infrastructure — regions, AZs, edge locations
- EC2 instance families, sizing and purchase options
- AMIs, user data and instance metadata
- S3 buckets, versioning, lifecycle and storage classes
- VPC, subnets, route tables, internet and NAT gateways
- Security groups and network ACLs
- IAM users, groups, roles and policy structure
- The AWS CLI, profiles and credential precedence
- CloudFormation as the baseline IaC tool
- Reading the AWS documentation efficiently

Module 3: Multi-AZ, Multi-Region & Disaster Recovery Patterns

Schedule: Week 2

The resilience material the exam tests hardest, framed the way its questions are framed: given an RTO and an RPO, which architecture meets them at the lowest cost. That pairing is the key — almost every resilience question is really a cost question with a constraint attached.

You work through the four documented DR strategies — backup and restore, pilot light, warm standby, multi-site active-active — and their real recovery times and price. Then Multi-AZ against read replicas and why they solve different problems, Route 53 failover and health checks, cross-region replication, and the difference between a highly available architecture and a fault-tolerant one, which the exam distinguishes and most candidates do not.

Topics

- RTO and RPO as the deciding constraints
- Backup and restore, pilot light, warm standby, active-active
- Multi-AZ versus read replicas — different problems
- Auto Scaling groups, health checks and lifecycle hooks
- Elastic Load Balancing — ALB, NLB and GWLB selection
- Route 53 routing policies and failover
- Cross-region replication for S3 and RDS
- Aurora global database and failover behaviour
- High availability versus fault tolerance
- Choosing the cheapest architecture that meets the target

Module 4: Domain 1 — Design Resilient Architectures (~26%)**Schedule: Weeks 2–3**

Resilience is the largest exam domain. Cover Multi-AZ vs Multi-Region patterns, Auto Scaling with target tracking and step scaling, Application Load Balancer + Network Load Balancer use cases, Route 53 routing policies (simple, weighted, latency, failover, geolocation, multi-value), RDS Multi-AZ vs read replicas vs Aurora Global, S3 replication (CRR, SRR, RTC), DynamoDB Global Tables, plus the disaster-recovery patterns (backup-restore, pilot light, warm standby, multi-site active-active). Domain-specific question drills throughout.

Topics

- Multi-AZ vs Multi-Region
- Auto Scaling — target tracking, step, scheduled
- ALB vs NLB vs Gateway Load Balancer
- Route 53 routing policies
- RDS Multi-AZ vs read replicas
- Aurora Global Database
- S3 replication — CRR, SRR, RTC
- DynamoDB Global Tables
- Disaster recovery patterns

Module 5: Decoupling — SQS, SNS, EventBridge & Step Functions**Schedule: Week 3**

The pattern the exam rewards more than any other. A question that describes a tightly coupled system with a failing component almost always has a queue in the correct answer, and recognising that shape quickly is worth several marks.

SQS standard against FIFO, visibility timeout, dead-letter queues and long polling; SNS fan-out and the SNS-to-SQS pattern; EventBridge for event-driven routing and scheduled rules; and Step Functions for orchestration where a state machine beats application code. Kinesis is covered for streaming, with the distinction from SQS made explicit, because the exam tests exactly that boundary.

Topics

- Recognising a coupling problem in a scenario
- SQS standard versus FIFO
- Visibility timeout, dead-letter queues and long polling
- SNS topics, subscriptions and fan-out
- The SNS-to-SQS fan-out pattern
- EventBridge rules, buses and scheduled events
- Step Functions for orchestration
- Kinesis Data Streams versus SQS
- Idempotency and duplicate delivery
- Asynchronous processing and buffering under load

Module 6: Domain 2 — Design High-Performing Architectures (~24%)

Schedule: Weeks 3–4

Performance domain. Cover compute selection (EC2 instance families and when each fits, Lambda performance tuning, ECS / EKS / Fargate trade-offs), storage selection (EBS volume types — gp3 / io2 / st1 / sc1, S3 storage classes and Intelligent-Tiering, EFS performance modes, FSx options), database performance (RDS instance scaling vs read replicas vs caching, DynamoDB capacity modes and DAX, ElastiCache Redis vs Memcached), plus content delivery and caching (CloudFront, Global Accelerator, ElastiCache). Domain-specific question drills throughout.

Topics

- EC2 instance families and selection
- Lambda performance tuning
- ECS / EKS / Fargate trade-offs
- EBS volume types
- S3 storage classes and Intelligent-Tiering
- RDS performance and read replicas
- DynamoDB capacity, DAX
- ElastiCache Redis vs Memcached
- CloudFront and Global Accelerator

Module 7: Storage Selection — S3 Classes, EBS, EFS & FSx

Schedule: Week 4

Storage questions are among the most reliably answerable on the exam, provided you know the selection criteria rather than the marketing. Block, file and object storage and which problems each is actually for; EBS volume types and the IOPS and throughput each delivers; EFS for shared POSIX access; and FSx for Windows and Lustre workloads.

S3 storage classes get the most time, including the retrieval charges and minimum durations that make Glacier Deep Archive the wrong answer for anything accessed monthly. Lifecycle policies, Intelligent-Tiering, Transfer Acceleration and Storage Gateway close the module, along with the migration services the exam expects you to distinguish.

Topics

- Block, file and object storage — choosing correctly
- EBS volume types, IOPS and throughput
- EBS snapshots, encryption and multi-attach
- Instance store and its trade-offs
- EFS performance and throughput modes
- FSx for Windows and FSx for Lustre
- S3 storage classes, retrieval cost and minimum duration
- Lifecycle policies and Intelligent-Tiering
- S3 Transfer Acceleration and multipart upload
- Storage Gateway, DataSync and Snow family

Module 8: Database Selection — RDS, Aurora, DynamoDB & Caching

Schedule: Week 4

The other reliably answerable family, and the one where a wrong choice in a scenario costs the whole question. Relational against non-relational and the access patterns that decide it; RDS engines, Multi-AZ, read replicas and backup behaviour; Aurora's storage architecture and why its failover characteristics differ.

DynamoDB gets proper treatment because the exam probes it: partition keys and hot partitions, on-demand against provisioned capacity, global and local secondary indexes, DynamoDB Streams and global tables. ElastiCache for Redis and Memcached covers caching strategies, and the module closes on the purpose-built services — Redshift, Neptune, DocumentDB, Timestream — at the level the exam actually tests, which is knowing which one a scenario is describing.

Topics

- Relational versus non-relational access patterns
- RDS engines, Multi-AZ and read replicas
- Automated backups, snapshots and point-in-time recovery
- Aurora architecture, replicas and failover
- DynamoDB partition keys and hot partitions
- On-demand versus provisioned capacity
- Global and local secondary indexes
- DynamoDB Streams and global tables
- ElastiCache Redis and Memcached, and caching strategies
- Redshift, Neptune, DocumentDB and Timestream — recognising the fit

Module 9: IAM in Depth — Policies, Roles, SCPs & Federation

Schedule: Week 5

The single largest scored domain rests on this, and it is where careless candidates lose marks they could have kept. Policy evaluation logic taught precisely — explicit deny beats everything, then organisation SCPs, then permission boundaries, then identity and resource policies — because the exam writes questions that turn on exactly this order.

Roles and `sts:AssumeRole` for cross-account access; instance profiles so an application never holds a key; resource-based policies and where they differ from identity-based; Organizations, service control policies and permission boundaries; and identity federation with SAML, Cognito and IAM Identity Center. The rule enforced throughout is least privilege, which is also the exam's default correct answer.

Topics

- Policy evaluation logic and explicit deny
- Identity-based versus resource-based policies
- Roles, `sts:AssumeRole` and cross-account access
- Instance profiles and never embedding keys
- Organizations, OUs and service control policies
- Permission boundaries and their interaction
- IAM Identity Center and SAML federation
- Cognito user pools and identity pools
- Least privilege and policy conditions
- Reading a policy and predicting the outcome

Module 10: Domain 3 — Design Secure Applications (~30%)

Schedule: Weeks 5–6

The largest exam domain. IAM in depth — policy evaluation logic, permission boundaries, conditions, IAM Access Analyzer. Network security — Security Groups, NACLs, VPC endpoints (Gateway and Interface), PrivateLink, Direct Connect, VPN. Data security — KMS (CMKs, key rotation, envelope encryption), Secrets Manager vs Parameter Store, ACM, S3

encryption options, RDS encryption. Plus AWS WAF, Shield Standard / Advanced, GuardDuty, Macie, Inspector, Security Hub, plus the compliance baselines (PCI DSS, HIPAA, SOC 2). Domain-specific question drills throughout.

Topics

- IAM policy evaluation, permission boundaries, conditions
- Security Groups and NACLs in depth
- VPC endpoints — Gateway and Interface
- PrivateLink, Direct Connect, VPN
- KMS — CMKs, rotation, envelope encryption
- Secrets Manager vs Parameter Store
- ACM and TLS termination
- WAF, Shield Standard / Advanced
- GuardDuty, Macie, Inspector, Security Hub
- Compliance baselines

Module 11: Data Protection — KMS, Secrets, and Encryption in Transit

Schedule: Week 6

The rest of the security domain, and the part that appears in scenario questions about compliance. KMS keys — AWS-managed, customer-managed and customer-provided — key policies, envelope encryption, rotation and multi-region keys, plus CloudHSM and the narrow set of cases that genuinely require it.

Encryption at rest across S3, EBS, RDS and DynamoDB; encryption in transit with ACM, TLS termination choices and where the exam expects it terminated. Secrets Manager against Parameter Store, including the rotation difference that is usually the deciding factor. The module closes on the detective controls — CloudTrail, Config, GuardDuty, Security Hub, Macie and WAF — at the level of knowing which one a scenario needs.

Topics

- KMS key types, key policies and grants
- Envelope encryption and data keys
- Key rotation and multi-region keys
- CloudHSM and when it is genuinely required
- Encryption at rest across S3, EBS, RDS and DynamoDB
- ACM, TLS termination and certificate management
- Secrets Manager versus Parameter Store
- CloudTrail, Config and AWS Organizations trails
- GuardDuty, Security Hub, Macie and Inspector
- WAF, Shield and protecting an edge

Module 12: Domain 4 — Design Cost-Optimised Architectures (~20%)

Schedule: Week 7

Cost optimisation. Compute pricing (on-demand, spot, savings plans, reserved instances), storage tiering (S3 Intelligent-Tiering, lifecycle rules, Glacier classes), data-transfer costs (the often-forgotten exam topic), plus cost-optimisation tooling (Cost Explorer, Trusted Advisor, AWS Budgets, Cost Anomaly Detection). Domain-specific question drills throughout. Plus the discipline of reading exam questions for cost-optimisation keywords ('most cost-effective', 'minimise cost').

Topics

- EC2 pricing — on-demand, spot, savings, reserved
- Spot strategies — Fleet, mixed instances
- S3 Intelligent-Tiering and lifecycle
- Glacier classes — Instant, Flexible, Deep Archive
- Data-transfer cost optimisation

- Cost Explorer and Cost Anomaly Detection
- AWS Budgets and Trusted Advisor

Module 13: Well-Architected Framework & Architecture Review

Schedule: Week 7

The framework the exam is written against, used as the review instrument it is meant to be. The six pillars — operational excellence, security, reliability, performance efficiency, cost optimisation and sustainability — and the design principles under each, which is where a surprising number of question stems come from directly.

You then run a review on your own architecture from earlier weeks: identify the risks, propose remediation, and state the trade-off each one costs. That exercise is the closest thing the course has to the actual job, and it is also the best preparation for the exam's recurring shape — several technically valid answers, one that best matches a stated priority.

Topics

- The six pillars and their design principles
- Trade-offs between pillars, stated explicitly
- Running a Well-Architected review
- Identifying and prioritising risks
- Cost optimisation as a design activity
- Right-sizing, Savings Plans and Reserved Instances
- Spot capacity and interruption-tolerant design
- Recognising the priority a question is testing
- Eliminating technically valid but wrong answers
- Documenting an architecture decision

Module 14: Mock Exam 1, Architecture Walkthroughs & Gap Closure

Schedule: Week 8

First full-length timed mock exam (65 questions, 130 minutes) under real exam conditions — including the Pearson VUE-style interface. Detailed answer review for every question, focusing on why the wrong options are wrong (the SAA-C03 grading model rewards understanding-of-trade-offs more than memorisation). Plus 5 architecture walkthroughs of common scenarios — three-tier web app, serverless event-driven app, BFSI compliance-aware setup, multi-region active-active, hybrid cloud — at whiteboard depth.

Topics

- Mock exam 1 — full 65 questions, 130 minutes timed
- Detailed answer review
- Architecture walkthroughs — 5 common scenarios
- Whiteboard scenario practice
- Gap-closure plan based on mock exam scores

Module 15: Mock Exam 2, Booking & Test-Day Prep

Schedule: Week 9 + 1 week buffer

Second full-length mock exam under real exam conditions — students who score 75–85% on this mock typically pass the real exam first attempt. Final gap-closure on weak domains, exam-booking guidance (Pearson VUE registration, ID requirements, online vs centre considerations), test-day mental-prep, plus a 30-minute strategy session on time management during the exam. Includes 'what to do if you don't pass on first attempt' coaching.

Topics

- Mock exam 2 — full 65 questions, 130 minutes timed
- Final gap-closure
- Pearson VUE booking — online vs test centre

- Test-day logistics and ID requirements
- Time-management strategy during the exam
- Retake strategy if first attempt fails

Module 16: AI-Assisted Development Workflow

Schedule: Final Week

The skill every 2026 hiring panel now probes for — building real work with AI in the loop, responsibly. Learn to drive AI assistants (GitHub Copilot, Claude, Cursor, and IDE-native AI) to scaffold and accelerate the tools and stack this course covers, generate tests, explain and refactor unfamiliar code, and cut the boilerplate — while keeping you firmly in control of every decision. Heavy focus on guardrails: reviewing each AI suggestion, spotting hallucinated APIs or wrong answers, and handling licensing and data-privacy concerns. Close with a mini-project that takes a deliverable end-to-end using an AI-assisted workflow, then fold the same tooling into version control and everyday team practice.

Topics

- AI assistants — GitHub Copilot, Claude, Cursor, IDE-native AI
- Effective prompting for this course's stack — scaffolding, boilerplate, config
- AI-assisted test generation and coverage
- Explaining, refactoring, and modernising unfamiliar code with AI
- AI-driven review, error detection, and quality checks
- Generating and maintaining documentation with AI
- AI debugging — interpreting errors, logs, and failing output
- Guardrails — reviewing output, avoiding hallucinations, licensing & data privacy
- Team workflow — AI in the editor, in reviews, and in delivery pipelines
- Mini-project — a deliverable built end-to-end with an AI-assisted workflow

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.